Introduction to Information Systems

Engineering Information Systems and Digital Skills

Lecture 1 | Introduction

First-year Engineering Students

Why are we studying Information Systems?

From using technology to understanding how information systems work

Engineers work inside organizations that depend on information systems.

Engineering decisions increasingly depend on data, software, communication networks, and digital services.

Information systems connect people, technology, data, and processes.

Understanding the system helps engineers use technology effectively—not simply use individual applications.

By the end of this lecture, you should be able to…

Define an Information System (IS) and distinguish IS from Information Technology (IT).

Identify the five major resources of an information system.

Explain the basic activities of an information system.

Recognize major types of information systems used by organizations.

Explain why information systems matter to engineers.

Identify ethical challenges associated with the use of information technology.

What is a System?

A foundation for understanding Information Systems

A system is a group of interrelated components working together toward a common goal.

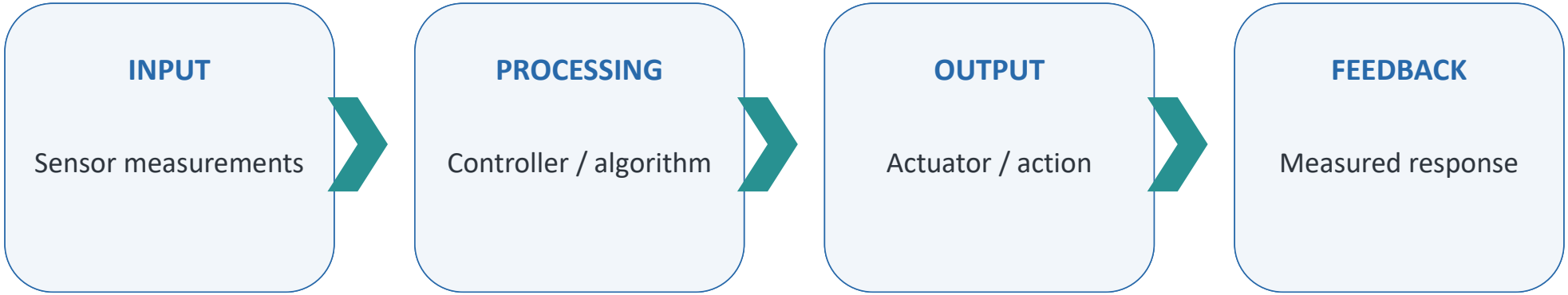
A system has a defined boundary and interacts with its environment.

Typical system activities include:

- Input — capturing and assembling elements that enter the system
- Processing — transforming inputs into outputs
- Output — delivering the results to their destination
- Feedback and control — monitoring performance and guiding the system

Think of an Engineering System

The same system logic appears in both physical and information systems



What is an Information System?

An organized combination of people, hardware, software, communications networks, and data resources that stores, retrieves, transforms, and disseminates information.

Key idea: an IS is more than a computer or a software application.

Information Systems vs. Information Technology

Information Systems (IS)

The complete set of resources and activities used to deliver information and information-processing functions.

Includes people, processes, hardware, software, data, and networks.

Focuses on how these elements work together.

Information Systems (IS) is a broader framework that combines that technology with people, data, and business processes to achieve organizational goals

Information Technology (IT)

Technology used to support information processing.

Includes hardware, software, networks, and related technologies.

IT is a major part of an information system—not the whole system.

Information Technology (IT) focuses on the physical tools and infrastructure (like hardware, software, and networks)

The Five Major IS Resources

PEOPLE

Users & specialists

HARDWARE

Physical devices

SOFTWARE

Instructions & programs

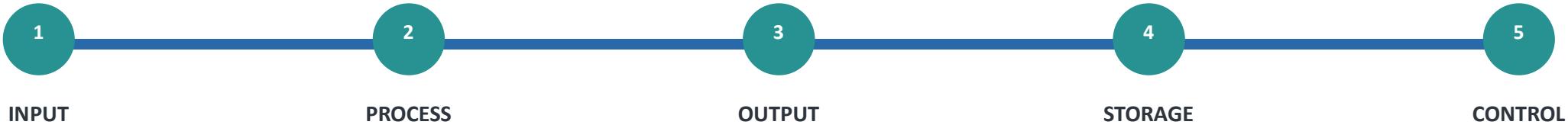
DATA

Facts & observations

NETWORKS

Communication resources

What does an Information System do?



Example: University registration → student data entered → system processes registration → timetable/records produced → data stored → rules and checks control the process.

Why should an Engineer understand Information Systems?

Engineering work is increasingly connected to digital information and networked systems.

Engineers use information systems for design, manufacturing, maintenance, project management, communication, and decision making.

Digital systems connect engineering teams with customers, suppliers, managers, and other professionals.

Understanding IS helps engineers communicate effectively with IT specialists and other organizational functions.

Major Types of Information Systems

Operations Support Systems

Transaction Processing Systems (TPS)

Process Control Systems (PCS)

Enterprise Collaboration Systems

Support day-to-day operations and transactions.

Management Support Systems

Management Information Systems (MIS)

Decision Support Systems (DSS)

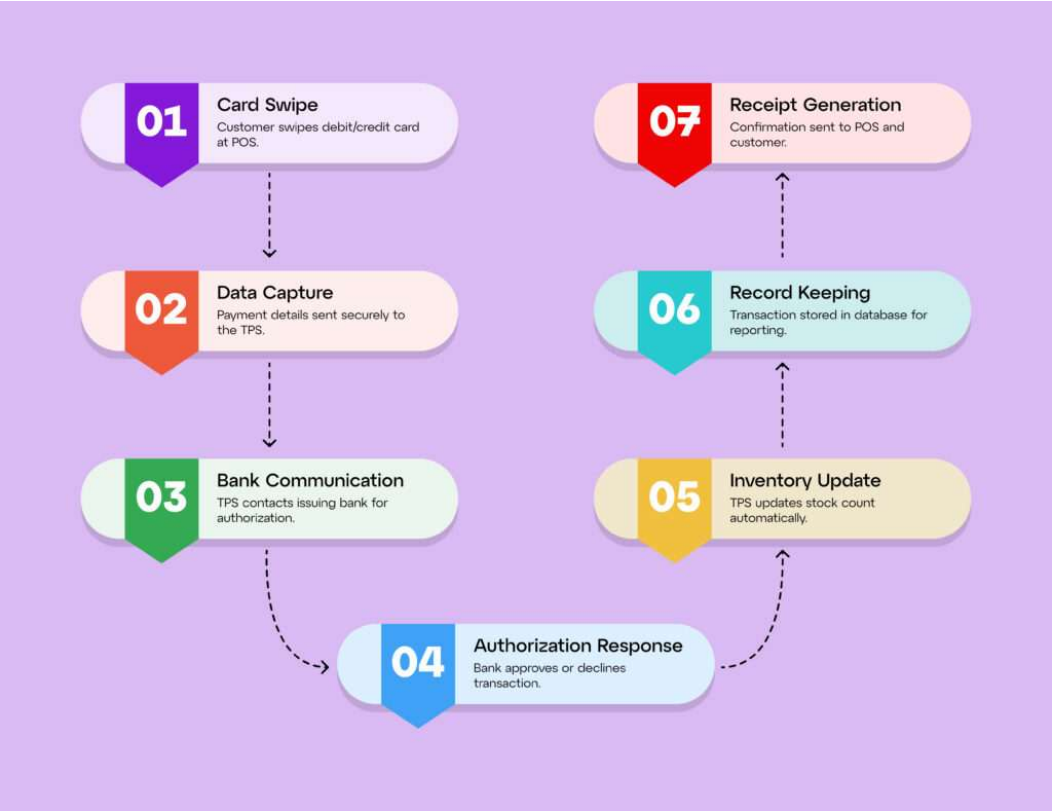
Executive Information Systems (EIS)

Support analysis, reporting, and decision making.

Operations Support Systems: Transaction Processing Systems (TPS)

A Transaction Processing System (TPS) is a software-based information system that collects, stores, modifies, and retrieves routine daily business transactions.

Point-of-Sale (POS) Systems: When you swipe a credit card at a retail store, the Point-of-Sale Systems authorize the payment with your bank, record the sale, and instantly lower the store's inventory count.



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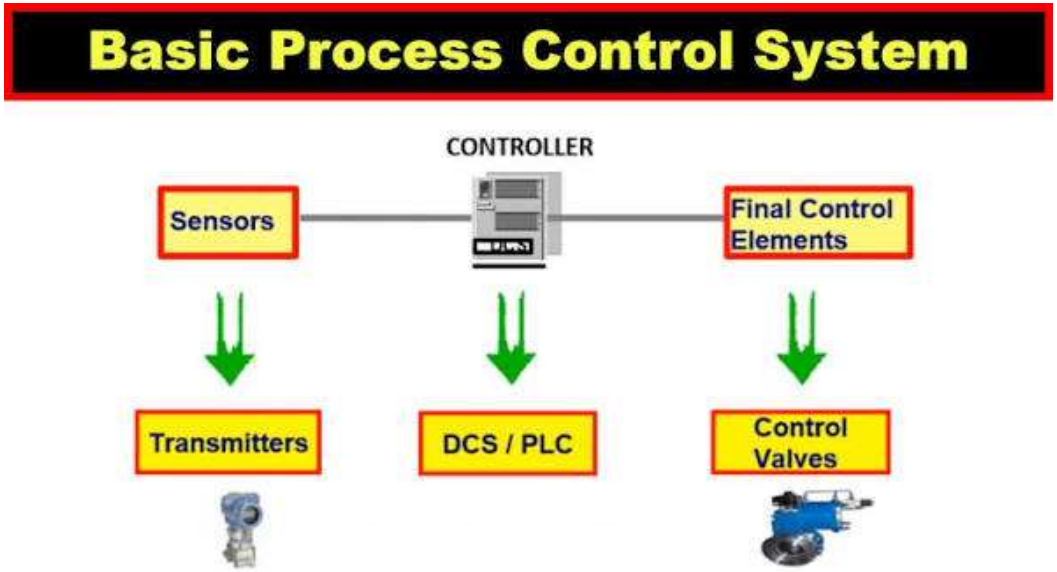
ATM (Automated Teller Machine) Withdrawals: When you withdraw cash, the ATM Transaction Delivery Systems verify your PIN, check your account balance, dispense the money, and update your bank records at the same time

Airline Reservation Systems: When you book a seat on a flight, the system locks that specific seat so it cannot be double-booked while processing your ticket payment and issuing an e-ticket.

Operations Support Systems: Process Control Systems (PCS)

A Process Control System (PCS) is an industrial automation network that monitors and regulates physical variables like temperature, pressure, and flow rates to keep manufacturing and production facilities running safely and efficiently

Tank Liquid Level Control: A level transmitter measures liquid volume in an industrial tank, a controller compares it to a target set point, and a control valve adjusts the inflow or outflow to maintain the desired level.



Operations Support Systems: Process Control Systems (PCS)

A Process Control System (PCS) is an industrial automation network that monitors and regulates physical variables like temperature, pressure, and flow rates to keep manufacturing and production facilities running safely and efficiently

Chemical Reactor Temperature Control: Temperature sensors monitor a chemical reaction inside a vessel, and the system automatically regulates cooling water or steam flow through a valve to keep the temperature steady.

Automotive Assembly and Painting: Programmable Logic Controllers (PLCs) coordinate robotic arms and conveyor speeds to weld frames, apply exact coats of paint, and assemble parts efficiently.

Management Support Systems: Management Information Systems (MIS)

Management Information Systems (MIS) combine business, technology, and people to collect, process, and analyze data for better decision-making

Core Real-World Example: SAP ERP

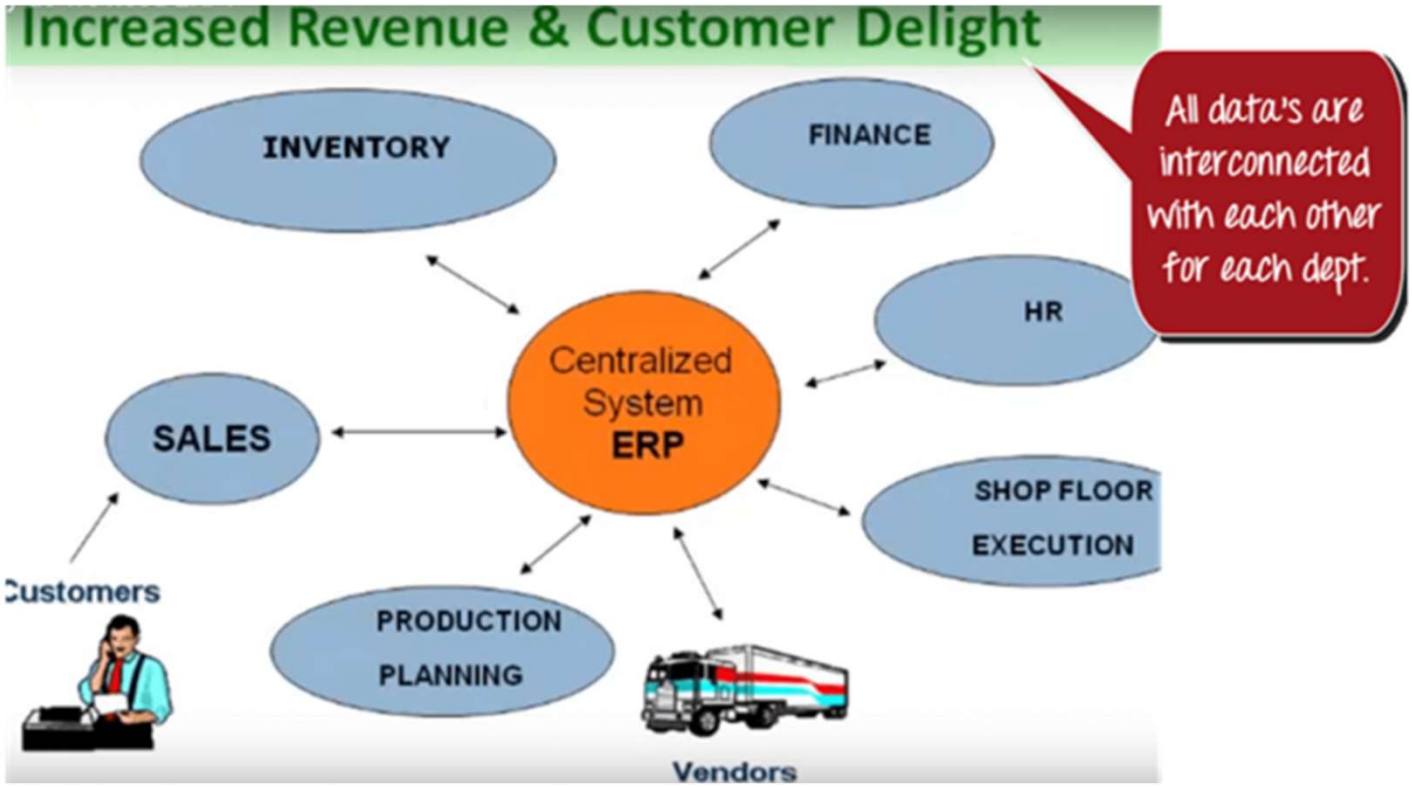
One of the most widely used enterprise examples of an MIS is SAP ERP (Enterprise Resource Planning). It acts as a centralized brain for a company by integrating multiple data streams:

Data Collection: The system automatically pulls transaction data, such as a cashier scanning an item at a register or a warehouse logging new stock.

Data Processing: Instead of showing thousands of raw, individual transactions, the system cleans and aggregates the data.

Managerial Output: The mid-level manager receives a structured Daily Sales & Inventory Report. If stock drops below a certain threshold, the system flags it so the manager can approve a reorder.

Management Support Systems: Management Information Systems (MIS)



Information Systems in Engineering

Manufacturing: production monitoring, quality data, inventory and process control.

Energy: monitoring, data collection, forecasting and asset management.

Mechatronics: sensors, controllers, embedded systems, networks and data.

Communications: network management, information exchange and digital services.

Project management: schedules, documents, collaboration and resource information.

Ethical Challenges in Information Technology

Privacy — who should have access to personal or organizational data?

Accuracy — how do we ensure that information is correct and reliable?

Intellectual property — who owns software, data, images, and digital content?

Security — how do we protect systems and information from misuse?

Responsible use — how should people behave when using organizational technology?

Lecture Summary

An Information System combines people, hardware, software, data, and networks.

IS transforms data into useful information through input, processing, output, storage, and control.

Information systems support both organizational operations and management decision making.

Engineers interact with information systems throughout their education and professional careers.

Technology must be used effectively, securely, and ethically.

30-Minute Interaction & Practice

Move from concepts to examples

Activity 1 — 10 min

In pairs, identify 3 information systems you use at university.

For each: people, hardware, software, data, networks.

What information does each system produce?

Activity 2 — 10 min

Choose one engineering system.

Identify its input, processing, output, storage and control.

Share one example with the class.

Activity 3 — 10 min: “IS or IT?” — classify 5 examples and explain your reasoning.

Quick Practice: What Is the Information System?

University LMS

Students submit assignments → system stores and processes submissions → grades/feedback are returned.

Smart Factory

Sensors collect production data → software analyzes it → operators receive alerts/actions.

Online Store

Customer places order → transaction is processed → payment/inventory/order information is updated.

Exit Ticket

Answer individually in 2–3 minutes

1. What is the difference between an Information System and Information Technology?
2. Name the five major IS resources.
3. Give one example of an Information System used by an engineer.
4. Identify one ethical issue related to information technology.
5. In one sentence: Why does an engineer need to understand Information Systems?

Next Lecture

Computer Hardware & Software

How the physical and software components support an Information System